

Quadrupled semitendinosus anterior cruciate ligament graft compared to a bone-patellar tendon bone approach: A systematic review and meta-analysis of functional outcomes

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ABSTRACT

Background: The choice of graft in anterior cruciate ligament (ACL) reconstruction influences postoperative pain, stability, and functional outcomes. Bone-patellar tendon-bone (BTB) autografts provide robust reconstructions but may increase anterior knee pain whereas quadrupled semitendinosus (ST) grafts may reduce morbidity while maintaining comparable function. However, direct comparisons of BTB and quadrupled ST remain limited. The primary objective of this systematic review and meta-analysis was to compare postoperative pain between BTB and quadrupled ST autografts. Secondary objectives were to evaluate instrumented stability (KT-1000), clinical stability (Lachman), patient-reported outcomes (Lysholm, Tegner, IKDC grades), and graft rupture rates.

Methods: Following PRISMA guidelines, MEDLINE/PubMed, EMBASE, SCOPUS, and Cochrane Central were searched through March 1, 2026, for randomized and comparative cohort studies of primary ACL reconstruction comparing BTB and quadrupled ST autografts. Eligible studies involved patients ≥ 16 years with or without meniscal or chondral pathology and ≥ 3 -month follow-up. Continuous and dichotomous outcomes were pooled as mean or standardized mean differences (SMD) and odds ratios (OR) with 95% confidence interval (CI) using random-effects models. Statistical heterogeneity was quantified using I^2 . Risk of bias was assessed using RoB 2 for randomized trials and ROBINS-I for nonrandomized comparative cohort studies.

Results: Ten studies were included ($n = 1029$), comprising 692 males (67%) and 337 females (33%). The mean ages across studies ranged from 17.7 to 49.4 years, with follow-up periods spanning 6 months to 8 years. ST reduced postoperative pain (SMD 0.44; 95% CI 0.09–0.80, $p = 0.015$), whereas BTB improved KT-1000 stability (SMD -0.70 ; 95% CI -1.02 to -0.38 , $p < 0.0001$). IKDC Grade A favored ST (OR 1.31; 95% CI 1.03–1.66, $p = 0.039$). Lachman, Lysholm, Tegner, and graft rupture did not differ between groups.

Conclusion: Both BTB and quadrupled ST autografts are effective for ACL reconstruction. BTB provides greater anterior stability, while ST is associated with less postoperative pain and higher odds of achieving IKDC Grade A outcomes without increasing failure risk. These findings highlight the importance of individualized graft selection to balance mechanical stability with patient-reported recovery and donor-site morbidity.

1. Introduction

Anterior cruciate ligament (ACL) reconstruction is a common procedure performed in young and active patients to restore knee stability and facilitate return to sport. Autograft selection remains a central

decision in ACL surgery as graft choice can influence early postoperative pain, donor-site morbidity, objective stability, and patient-perceived recovery.¹ Commonly used autograft options include bone-patellar tendon-bone (BTB), combined semitendinosus-gracilis (STG), and autologous semitendinosus (ST) grafts.^{2,3} The BTB graft

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has been the “gold-standard” graft for ACL reconstruction for several years; however, undesirable sequelae such as anterior knee pain, and other donor site morbidities has led to the development and increased utilization of hamstring-based grafts as a primary alternative.⁴ Many comparative studies have demonstrated that hamstring grafts, particularly the STG graft, reduces these adverse outcomes. However, overall knee stability and functional outcomes are generally equivalent between STG and BTB.⁵

By sparing the gracilis tendon, the ST graft may offer distinct advantages over the STG and BTB grafts. Firstly, preservation of the gracilis minimizes donor site morbidity, including reduced pain and fewer postoperative complications, which can improve early comfort and rehabilitation.⁶ Secondly, the gracilis tendon contributes to knee flexion and internal rotation, movements that are particularly relevant for athletes requiring agility and pivoting.⁷ Thus by preserving the gracilis, the ST graft reduces knee flexor strength deficits, especially during the early postoperative period, without compromising functional outcomes.^{8,9} Matteucci et al. (2024) found that patients with ST autografts exhibited less knee flexor strength deficit at six months postoperatively compared to those with STG autografts.¹⁰ Finally, anterior knee pain is a well-recognized concern following BTB harvest and has been reported in a substantial proportion of patients, making donor-site symptom mitigation a clinically meaningful factor in graft selection.¹¹

From a construct standpoint, ST graft preparation typically involves harvesting the semitendinosus tendon and tripling or quadrupling (also known as a four strand).^{4,8} When prepared in this manner, the quadrupled ST graft achieves comparable tensile strength and thickness to that of the STG graft.^{4,8} Quadrupled ST grafts are also commonly paired with suspensory fixation strategies and may be used with socket-based (“all-inside”) ACL reconstruction approaches that create femoral and tibial sockets rather than full-length tunnels, often using retrograde drilling/reaming systems.^{12,13} However, surgical technique such as full-tunnel versus socket-based, varies across the ACL literature, and graft choice and technique are not always interchangeable; therefore, comparisons focused on graft type should be interpreted in the context of technique heterogeneity.^{4,5,8,12,13}

Much of the current literature has compared outcomes of BTB and STG grafts.^{2,3,14} However, there is more limited evidence directly comparing BTB to a quadrupled ST graft, particularly with respect to postoperative pain, graft stability, failure rates, and patient-reported functional outcomes. Given the differing morbidity profiles and biomechanical considerations of BTB versus ST constructs, as well as the clinical importance of pain and early recovery, greater clarity is needed regarding the comparative performance of BTB and quadrupled ST autografts.

Accordingly, this systematic review and meta-analysis aimed to compare postoperative pain (primary outcome) between BTB and quadrupled ST autografts. Secondary outcomes included stability, patient-reported functional outcomes and graft rupture rates.^{15–17}

2. Methodology

2.1. Search strategy

This systematic review and meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 reporting guidelines.¹⁸ The study is registered on PROSPERO under registration number CRD42021246045.

The following databases were searched: MEDLINE, PubMed, EMBASE, SCOPUS, and the Cochrane Central Register of Controlled Trials from database inception to March 1, 2026. The complete MeSH terms and search strategy for each database are provided in [Appendix 1](#).

Two reviewers (X.X and X.X) independently conducted title and abstract screening followed by full-text review to determine study eligibility. Disagreements were resolved by discussion and consensus, with adjudication by a third reviewer (X.X.X) when required. Inter-

reviewer agreement was quantified using Cohen's kappa (κ).

2.2. Selection criteria

Eligible studies included comparative randomized controlled trials and comparative cohort studies evaluating outcomes following primary ACL reconstruction using either bone-patellar tendon bone (BTB) autograft or a four-strand semitendinosus (4-ST) hamstring autograft.

Inclusion criteria were: 1) randomized studies and cohort studies, 2) primary autologous ACL reconstruction with BTB or quadrupled ST autografts, 3) patients 16 years and older with or without meniscal and chondral pathology. Concomitant meniscal and chondral pathologies being treated at the time of ACL reconstruction at the discretion of the study surgeon and minimum follow-up of 3 months were included. Specific treatment of the pathologies was not assessed.

Studies were excluded if they: 1) did not include a direct comparison between BTB and a semitendinosus-based hamstring autograft group; 2) were non-comparative designs (case series or case reports), review articles, editorials, conference abstracts without sufficient extractable data, or biomechanical/cadaveric studies; 3) evaluated revision ACL reconstruction or reported outcomes exclusively after revision procedures; or 4) had patients with prior ipsilateral knee surgery before ACL reconstruction, revision surgery after ACL reconstruction, concomitant ligamentous injury at the time of ACL rupture, and 5) patients who were pregnant or nursing at the time of study enrolment.

2.3. Outcome measures and data extraction

The primary outcome was postoperative knee pain using the validated Visual Analog Scale (VAS) scores.¹⁹ Studies reporting VAS pain scores at any postoperative timepoint were eligible for inclusion in the primary meta-analysis. Secondary outcomes included: 1) knee laxity (manual Lachman test and the KT-1000); 2) patient-reported outcomes of symptoms (Lysholm score); 3) return to physical activities (Tegner activity score); 4) overall knee function (IKDC grade); and 5) graft rupture.^{15–17,20,21} At least one outcome of interest was required for study inclusion. A single post-operative timepoint was used in meta-analysis calculation for each study due to variation in follow-up duration.

2.4. Meta-analysis methodology

Meta-analysis was performed using R (R version 4.2.0).¹⁴ To be able to compare outcomes across studies, a validated equation was used to convert medians and ranges to means and standard deviations, respectively.²² Standardized mean differences (SMDs) were used for continuous outcomes where studies used different scales, such as VAS scales that differed across studies. For continuous outcomes reported on the same scale, mean differences were used where applicable. Continuous outcomes were calculated as BTB minus ST such that negative values indicate lower scores in BTB than ST, and positive values indicate higher scores in BTB than ST. For dichotomous outcomes, odds ratios (ORs) were calculated when pooling was feasible.

Given expected clinical and methodological heterogeneity across studies due to differences in operative technique, fixation methods, and follow-up duration, a random-effects model was used for pooled analyses with 95% confidence interval (CI) reported. Specifically, due to variations in follow-up duration, a single post-operative score was used in the analysis. If multiple timepoints were available, the latest timepoint was included as this approach prioritizes clinically relevant longer-term outcomes while avoiding non-independence from multiple timepoints within the same cohort.

Statistical heterogeneity was assessed using the Chi-square test and quantified with the I^2 statistic.²³ The value of $I^2 < 25\%$ was considered low statistical heterogeneity, $I^2 25\text{--}49\%$, moderate statistical heterogeneity; I^2 greater than or equal to 50% was high statistical

heterogeneity.²³ Funnel plots were not performed as less than 10 studies contributed to any single outcome.

2.5. Methodological quality assessment

Risk of bias was assessed at the study level. Randomized controlled trials were evaluated using the Cochrane Risk of Bias 2 (RoB 2) tool, while observational cohort studies were assessed using the Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) tool.^{24,25} Assessments were performed independently by two reviewers, with disagreements resolved by consensus.

2.6. Certainty of evidence (GRADE)

The certainty of evidence for each main outcome (VAS pain, KT-1000, IKDC Grade A, and graft rupture) was assessed using the GRADE approach, considering risk of bias, inconsistency, indirectness, imprecision, and publication bias. Evidence from randomized trials was initially rated as high certainty and evidence from observational studies as low certainty, with downgrading or upgrading applied based on prespecified GRADE domains. Final certainty ratings were categorized as high, moderate, low, or very low.

3. Results

3.1. Study selection

Study selection is summarized in the PRISMA flow diagram (Fig. 1).

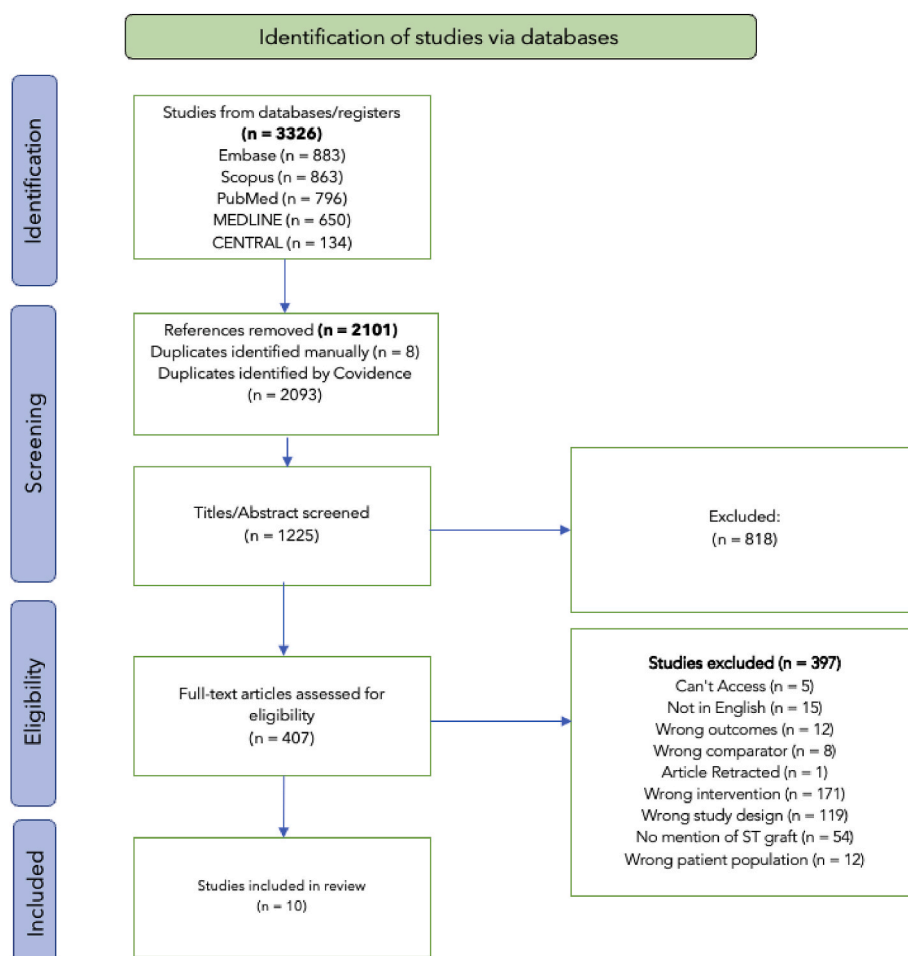


Fig. 1. PRISMA flow diagram for included studies.

Inter-reviewer agreement between reviewers was substantial at title/abstract screening ($\kappa = 0.82$) and substantial at full-text review ($\kappa = 0.76$) Risk of bias assessment can be seen in Figs. 2 and 3.

3.2. Study characteristics

This meta-analysis included a total of 10 studies, comprising three randomized controlled trials (RCTs) and seven cohort studies, with a sample size of 1029 patients. The mean ages of the patients across all included studies ranged from 17.4 to 49.4 years. Follow-up periods varied across studies, ranging from six months to eight years. Study-level characteristics and outcome measures are summarized in Table 1, and baseline demographics are summarized in Table 2. BTB grafts were found to be primarily secured via interference screws or press-fit techniques.^{26–28} ST grafts employed diverse methods including cortical suspension devices and tibial interferences screws.^{29–31}

3.3. Post-operative pain (VAS)

Four studies reporting postoperative VAS pain were included.^{26,29,32,33} Pooled analysis showed significantly lower postoperative pain in the semitendinosus (ST) group (SMD 0.44, 95% CI 0.09–0.80; $p = 0.015$), with substantial heterogeneity ($I^2 = 65.5\%$) (Fig. 4). In the acute postoperative period, Gobbi et al. (2004) reported a mean VAS score of 7.0 in the BTB group compared to 5.0 in the ST group.²⁷ At 3-year follow-up, the same study reported anterior kneeling pain in 15% of BTB patients (mean VAS 6.0) versus 7% of ST patients (mean VAS 4.0).²⁷ Eriksson et al. (2001) found that pain on kneeling was

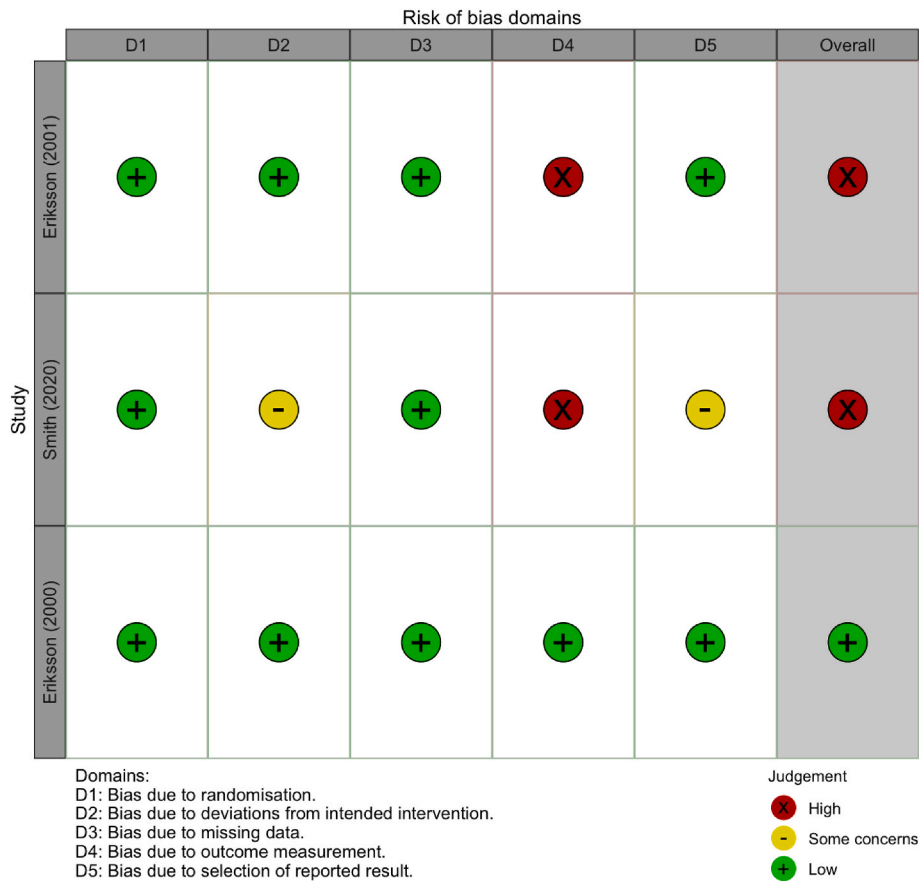


Fig. 2. Risk of bias analysis using the RoB 2 tool.

significantly more common in the BTB group (53%) compared to the ST group (16%) ($p < 0.01$). However, the same study noted that for general walking activity, there was no significant difference in VAS scores between groups at final follow-up ($p > 0.05$).²⁶

3.4. Clinical stability (Lachman score)

Lachman grades showed no statistically significant differences between graft types. Grade 0: OR 0.76 (95% CI 0.06–9.55; $p = 0.69$; $I^2 = 76.4\%$), grade 1: OR 1.32 (95% CI 0.12–15.09; $p = 0.67$; $I^2 = 74.7\%$), grade 2: OR 1.53 (95% CI 0.002–982.22; $p = 0.56$; $I^2 = 0\%$), and grade 3: OR 1.51 (95% CI 0.03–78.14; $p = 0.84$) (Fig. 5).

3.5. Instrumented stability (KT-1000)

Two studies contributed KT-1000 arthrometer data.^{29,30} Pooled analysis demonstrated significantly better anterior stability in the BTB group (SMD -0.70 , 95% CI -1.02 to -0.38 ; $p < 0.0001$), with no detected heterogeneity ($I^2 = 0.0\%$) (Fig. 4). Side-to-side laxity differences were assessed at maximum manual force at 134 N. Struwer et al. (2013) reported a mean side-to-side difference of 1.3 mm for BTB versus 2.4 mm for ST. Similarly, Pautasso et al. (2020) reported 2.5 mm for BTB versus 3.1 mm for ST.

3.6. Patient-reported outcomes

Lysholm (Post-operative): Four studies contributed postoperative Lysholm scores.^{30,32–34} There was no statistically significant difference between groups (SMD -0.24 , 95% CI -0.49 to 0.004 ; $p = 0.054$; $I^2 = 35.5\%$) (Fig. 4).

Tegner (Post-operative): Three studies reported postoperative

Tegner activity scores.^{30,32,33} No between-group difference was observed (SMD -0.02 , 95% CI -0.28 to 0.25 ; $p = 0.91$; $I^2 = 32.47\%$) (Fig. 4).

IKDC Grade (Postoperative): Three studies reported postoperative IKDC grades.^{26,28,32} The odds of achieving IKDC Grade A were higher in the ST group (OR 1.31, 95% CI 1.03–1.66; $p = 0.039$; $I^2 = 0.0\%$). No significant differences were observed for Grade B (OR 0.85, 95% CI 0.51–1.42; $p = 0.30$), Grade C (OR 1.10, 95% CI 0.44–2.76; $p = 0.69$), or Grade D (OR 0.96, 95% CI 0.60–1.56; $p = 0.78$) (Fig. 6).

3.7. Graft rupture

Four studies reported graft rupture outcomes.^{26,29,33,35} There was no significant difference in graft rupture between BTB and ST grafts (OR 0.88, 95% CI 0.38–2.01; $p = 0.76$; $I^2 = 0.0\%$) (Fig. 7). Absolute rupture rates varied between studies, ranging from 0 to 1.6% in the BTB group and 0–7.4% in the ST group.^{26,29,33,35}

3.8. Muscle strength

Postoperative muscle strength was reported in three studies.^{26,27,31} At 6 months postoperatively, Gobbi et al. (2004) reported a quadriceps work deficit of 23% in the BTB group compared to 7.3% in the ST group. Conversely, the ST group demonstrated a flexor work deficit of 22.4% compared to 21.3% in the BTB group. At 12 months, significant flexion deficits persisted in the ST group at $60^\circ/s$ and $180^\circ/s$ ($p < 0.05$).²⁷ At 2-year follow-up, Tow et al. (2005) found no significant difference in peak torque for flexion or extension between groups, with 4% deficit in the BTB group versus 9% in the ST group ($p > 0.05$).³¹ Similarly, Eriksson et al. (2001) reported no significant differences in functional strength, with both groups achieving a one-leg hop symmetry index of

Table 1

Summary of all included studies and outcome measures.

Author	Study Type	Intervention	Comparator (s)	Location of Study	Minimum Follow-Up	Outcome(s)
Smith et al. 2020 ¹⁸	Randomized Controlled Trial	4 ST	BTB	Columbia, Missouri, United States	24 months	KT-1000, IKDC Evaluation Form, Marx Activity, KOOS, IKDC Subjective, Patient reported pain Visual Analogue Scale (VAS), SF-12
Eriksson et al. 2001 ¹⁹	Prospective Clinical Study	4 ST	BTB	Lysholm, Sweden	24 months	Lysholm, Tegner activity scale, Visual Analog Scale (VAS) for knee function and activity level, range of motion, Lachman test, pivot shift, KT-1000 arthrometer, thigh circumference, 1-legged Hop Test, patellofemoral pain/crepitus, donor site morbidity
Pautasso et al. 2020 ²⁰	Retrospective Cohort Study	4 ST	BTB	Turin, Italy	24 months	KOOS, Lysholm, Tegner activity scale, KT-1000,
Struwer et al. 2013 ²¹	Retrospective Cohort Study	4 ST	BTB	Marburg, Germany	28 months	IKDC, Lachman, Lysholm, Tegner activity scale, Kellgren-Lawrence classification
Eriksson et al. 2000 ²²	Prospective, Randomized Controlled Trial	4 ST	BTB	Stockholm, Sweden	20 weeks	Laxity, One-leg hop test, Tegner Activity Level, Lysholm, Visual Analogue Scale (VAS), IKDC, Extension Deficit, Meniscal Pathology, Timing of Surgery
Boonriong & Kietsiroje, 2004 ²³	Prospective, Randomized Controlled Trial (RCT)	4 ST	BTB	Songkla, Thailand	1 year	Lachman, Pivot shift, Lysholm, Hospital for Special Surgery (HSS) knee score, Visual Analog Scale (VAS)
Gobbi et al. 2003 ²⁴	Prospective Clinical Study	4 ST	BTB	Milan, Italy	3 months	Surgical time, Postoperative pain, IKDC, Lysholm, Tegner activity scale, Visual Analog Scale (VAS), Noyes scale, isokinetic testing, computerized laxity analysis, Hop test, Kneeling Pain, Tunnel Widening, Range of Motion, Return to Sport, Pain Over Metal Hardware, Anterior Knee Pain
Murgier et al. 2021 ²⁵	Observational Cohort Study	4 STG, 4 ST, 5-6 STG, 7-8 STG	BTB	New Zealand	2 years	Graft failure rate
Gobbi A et al. 2004 ²⁶	Prospective Comparative Clinical Study	4 ST	BTB	Milan, Italy	6 months	IKDC, Single Assessment Numeric Evaluation (SANE) score, Range of motion, Patellofemoral joint examination (crepitus, pain, swelling, tightness), Visual Analog Scale (VAS), Knee laxity, Isokinetic muscle strength testing, Maximum peak torque and total work in flexion, Ergo jump test, Tegner activity scale, Noyes scale, Lysholm knee score, Return to sport, Graft dimensions, Surgical time
Tow et al. 2005 ²⁷	Prospective Clinical Study	4 ST	BTB	Singapore	24 months	Postoperative pain (1 day after surgery), Complications and additional surgeries, Gender differences in outcomes for each graft type
						KT1000, IKDC, Anterior Knee Pain, Muscle Strength and Endurance, Graft Failure Rates

Table 2

Summary of patient demographics in included studies.

Study	Total N	ST Group (Male/Female)	BTB Group (Male/Female)	Mean Age	Follow-Up
Murgier et al. 2021	441	261 (NR)	180 (NR)	<20*	24 mo
Eriksson et al. 2001 (JBJS)	164	84 (NR)	80 (NR)	25.7	31 mo
Pautasso et al. 2020	104	51 (NR)	53 (NR)	15–30	24 mo
Gobbi et al. 2004	80	40 (22 M/18 F)	40 (26 M/14 F)	28.0	36 mo
Boonriong & Kietsiroje 2004	75	30 (NR)	45 (NR)	27.0	12 mo
Tow et al. 2005	68	34 (32 M/2 F)	34 (32 M/2 F)	27.3	24 mo
Smith et al. 2020	56	27 (NR)	29 (NR)	17.7	24 mo
Struwer et al. 2013	41	22 (13 M/9 F)	19 (10 M/9 F)	49.4	32 mo
Eriksson et al. 2001 (Scand)	(107)	57 (NR)	50 (NR)	26.8	6 mo
Gobbi et al. 2003	(80)	40 (NR)	40 (NR)	28.0	36 mo
TOTAL	1029	549	480	17–49	

98% at a median follow-up of 31 months ($p = 0.8$).²⁶

3.9. Risk of bias assessment

Three randomized controlled trials were assessed using the RoB 2 and were judged to have low risk of bias.^{26,29,33} The remaining seven observational studies were assessed using ROBINS-I and were rated as moderate risk of bias, largely due to nonrandomized allocation and potential confounding.^{28,30–32,34–36} Please see Figure 2 and 3.

3.10. Certainty of evidence (GRADE)

Using GRADE, certainty of evidence was rated across the main outcomes (VAS pain, KT-1000, IKDC Grade A, graft rupture). Certainty was low for VAS pain due to inconsistency (substantial heterogeneity) and study design limitations (predominantly nonrandomized comparisons). Certainty was low for KT-1000 due to imprecision and limited contributing studies. Certainty was low for IKDC Grade A because relatively few studies contributed data and residual confounding is possible in non-randomized cohorts. Certainty was very low for graft rupture due to low event rates and wide confidence intervals, despite consistency in direction of effect.

4. Discussion

This study compared quadrupled semitendinosus (ST) and bone-patellar tendon-bone (BTB) grafts in ACL reconstruction. Pooled VAS pain favored the ST graft, demonstrating a significant reduction in postoperative pain. However, BTB grafts provided superior anterior knee stability, represented by the KT-1000 scores which were

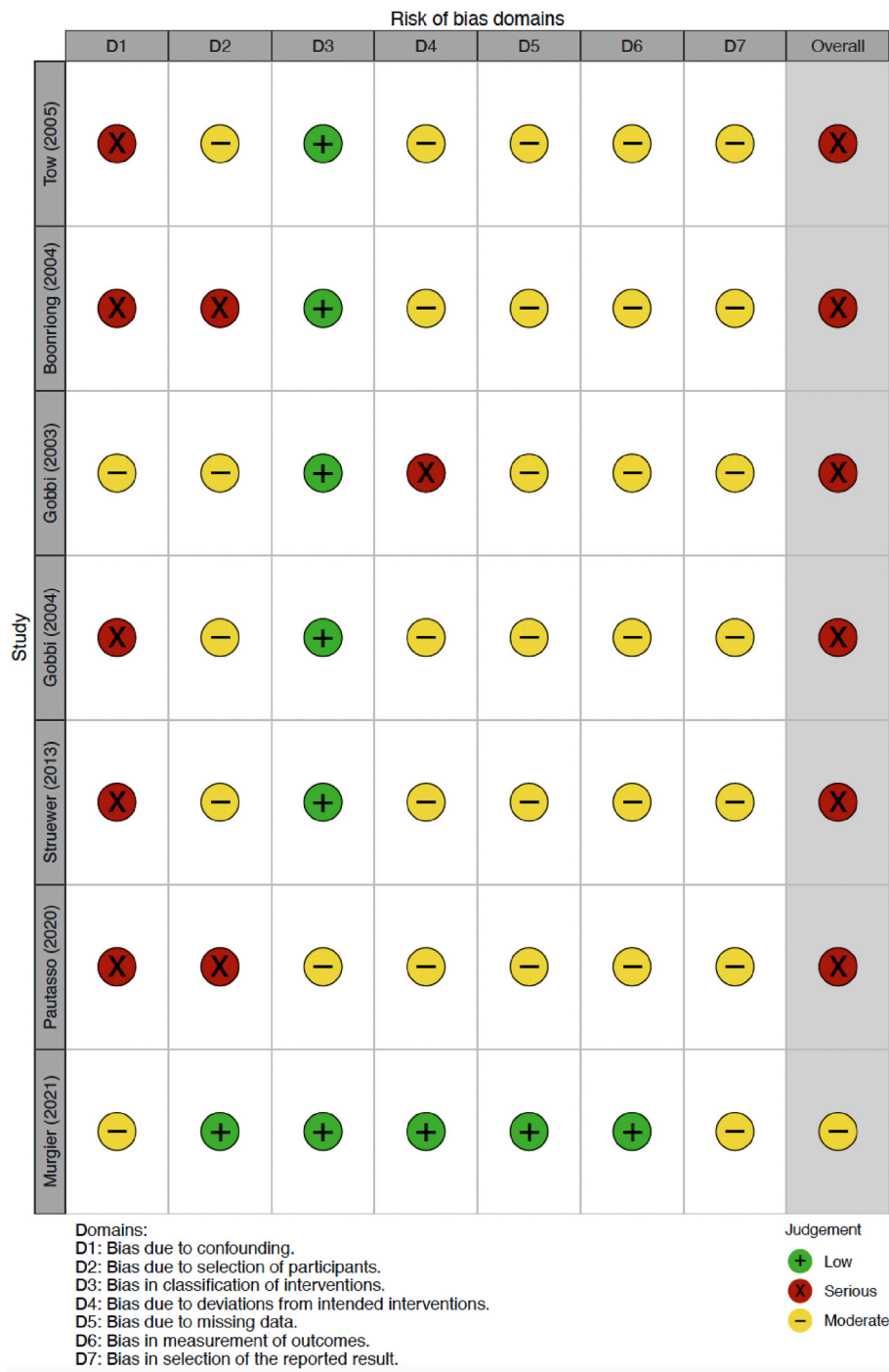


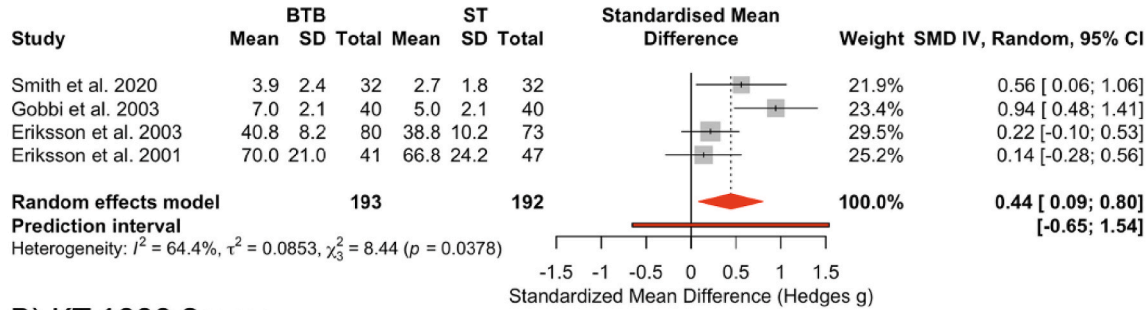
Fig. 3. Risk of bias analysis using ROBINS-I Tool.

significantly improved amongst the BTB grafts when compared to the ST grafts. Lachman grades, Tegner activity level and graft rupture rates were similar between quadrupled ST and BTB grafts. These results are broadly consistent with prior meta-analyses comparing BTB and hamstring autografts overall, while specifically addressing semitendinosus-only constructs rather than traditional semitendinosus–gracilis harvest.^{2,37}

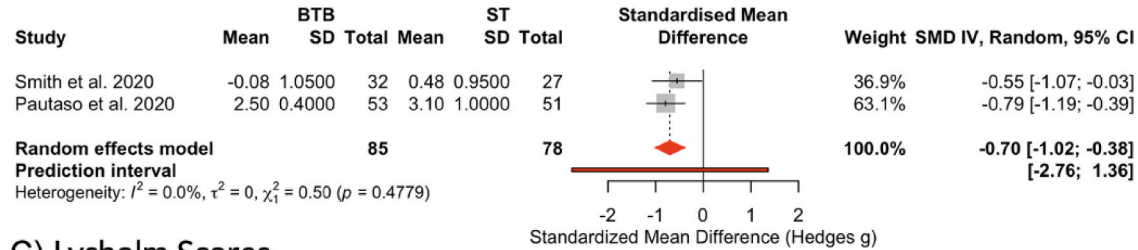
The pain advantage observed with the ST graft is clinically plausible given differences in harvest morbidity. Patellar tendon harvest can be associated with anterior knee symptoms that may influence early rehabilitation and patient comfort, whereas hamstring harvest tends to shift morbidity toward posterior thigh symptoms.^{37–39} It is worth noting

that donor site management varied across studies; for instance, Operative technique is an important consideration when interpreting postoperative pain and morbidity in semitendinosus-only reconstruction. ST-only grafts are frequently performed using “all-inside” approaches with socket-based tunnel preparation, and technical descriptions have proposed this approach as a method to reduce surgical morbidity and potentially improve early postoperative comfort.^{12,13} Several comparative cohorts have evaluated BTB versus all-inside ST constructs.^{8,13,40,41} In addition, morbidity profiles may differ when comparing ST-only constructs with traditional semitendinosus–gracilis harvest, which remains a common comparator in the broader ACL literature.⁴² Because included studies variably reported tunnel

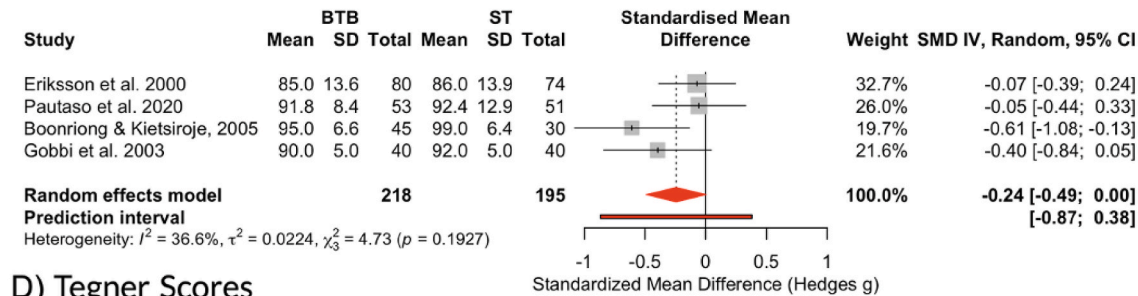
A) Visual Analogue Score (VAS)



B) KT-1000 Scores



C) Lysholm Scores



D) Tegner Scores

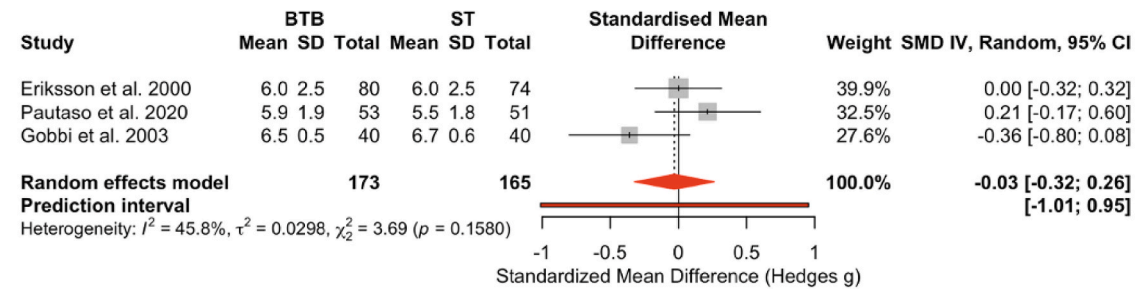


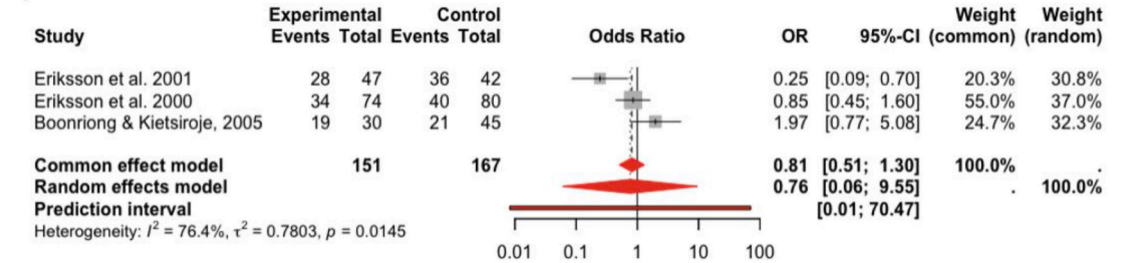
Fig. 4. Post-op forest plot Visual Analogue Scale pain scores, KT-1000 scores, Lysholm scores, and Tegner scores.

preparation, fixation, graft preparation, and rehabilitation details, we could not reliably isolate the extent to which the observed pain differences were attributable to graft choice alone versus technique-related factors (including “all-inside” methods).

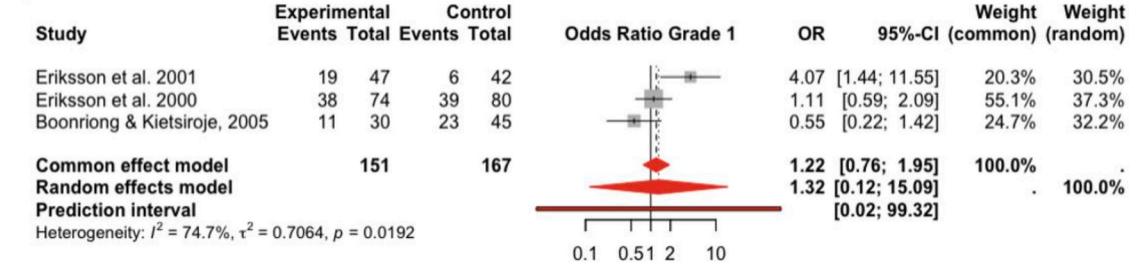
For patient-reported function, pooled Lysholm outcomes showed a small, borderline difference and Tegner activity did not. In contrast, IKDC Grade A was more frequent in the ST group. One interpretation is that categorical IKDC grading may capture symptom and function domains that are sensitive to donor-site morbidity or early recovery trajectories in a way that some continuous scores do not. More broadly, comparative literature has often reported largely similar functional outcomes between patellar tendon and hamstring grafts, with differences tending to cluster around morbidity and specific stability measures rather than global function.^{14,43} Within our included evidence in this analysis, improvements in subjective knee function have been reported in some cohorts, reinforcing that patient-perceived outcomes may not correlate one-to-one onto instrumented laxity.²⁶

Objective stability results favored BTB on KT-1000. This aligns with the concept that graft construct, fixation, and healing characteristics can influence device-based anterior translation measurements, and multiple included cohorts reported differences in instrumented laxity between grafts.^{9,27,33,44} However, the clinical relevance of this statistical difference warrants careful interpretation. The mean side-to-side laxity difference for BTB grafts (1.3–2.5 mm) consistently fell within the widely accepted threshold of <3 mm.⁴⁵ In contrast, the ST group in one included study exhibited a mean difference of 3.1 mm, slightly exceeding this clinical threshold.³⁰ This finding suggests that while both grafts provide adequate stability, the BTB construct may offer a “tighter” healing interface that provides a greater margin of safety against residual laxity.⁴⁶ Importantly, Lachman grading, which reflects the clinical examination rather than device-based translation, did not differ significantly across grades, suggesting that improved KT-1000 values with BTB may not necessarily translate into detectable differences on standard clinical examination within the time periods studied. In

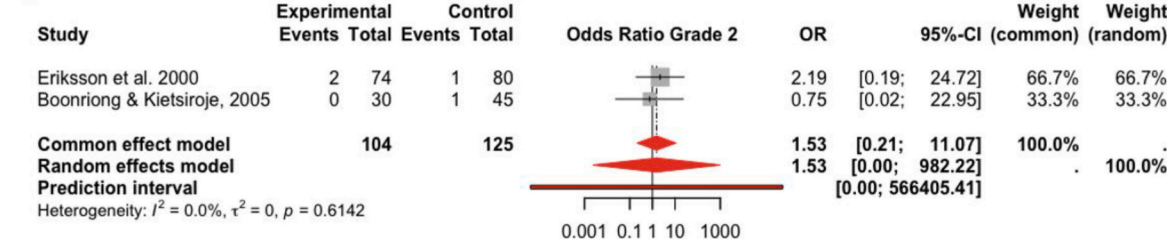
A) Grade 0



B) Grade 1



C) Grade 2



D) Grade 3

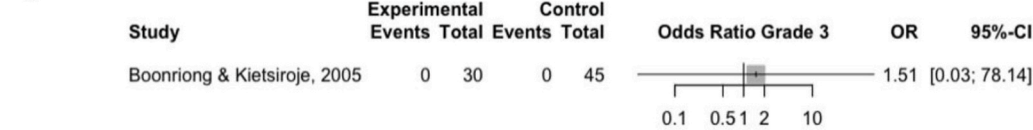


Fig. 5. Post-op forest plot Lachman Grade 0 to 3 respectively.

addition, the KT-1000 finding was based on a limited number of studies, which strengthens internal consistency but limits generalizability across broader patient populations and surgical practices.

This review does present with some limitations. The number of included studies was small for several outcomes (including KT-1000 and IKDC), and study designs ranged from randomized and prospective comparisons to retrospective cohorts, increasing susceptibility to selection bias and unmeasured confounding. Pain and some PROMs showed between-study heterogeneity and are sensitive to follow-up timing and perioperative protocols, which were variably reported. Finally, incomplete reporting of surgical technique and rehabilitation limited planned subgroup analyses, particularly for disentangling graft effects from “all-inside” technique effects in semitendinosus-only reconstruction.

5. Conclusion

Current evidence suggests that both BTB and semitendinosus-only autografts provide comparable clinical stability on examination and similar activity and rupture outcomes, with ST associated with less postoperative pain and BTB associated with less objective anterior laxity. Graft selection should therefore remain individualized, incorporating patient priorities and surgeon experience, while recognizing that operative technique and rehabilitation may modify graft-associated

trade-offs.

Guardian/patient consent

Informed written consent was not required for this study.

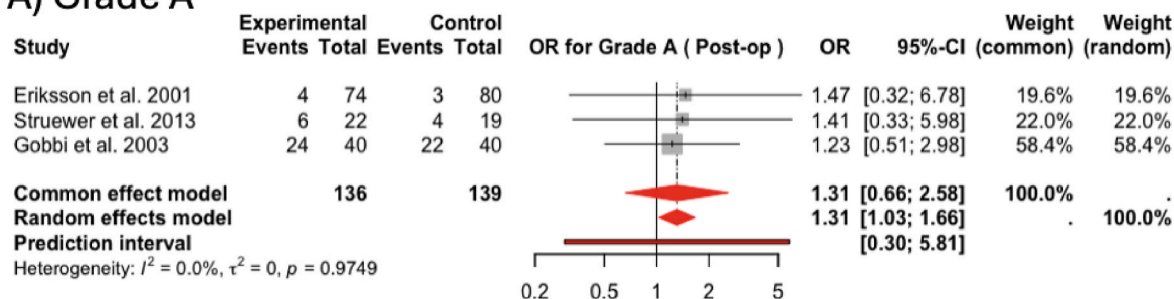
Ethical statement

This manuscript was produced without artificial intelligence software. None of the authors for this study had any conflicts of interests. The study is registered on PROSPERO under registration number CRD42021246045.

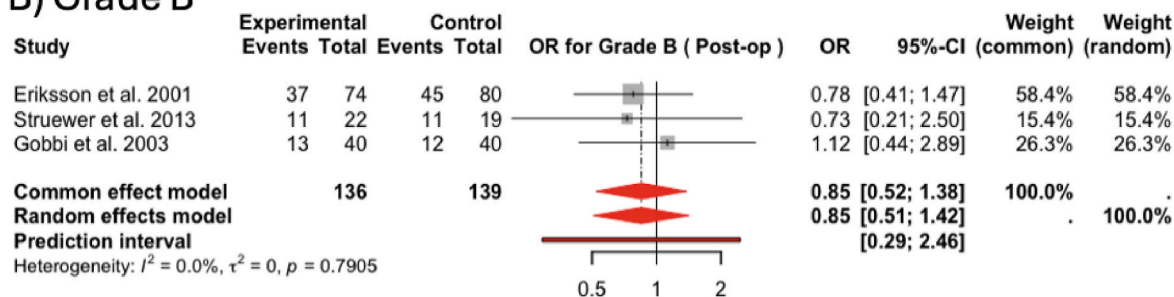
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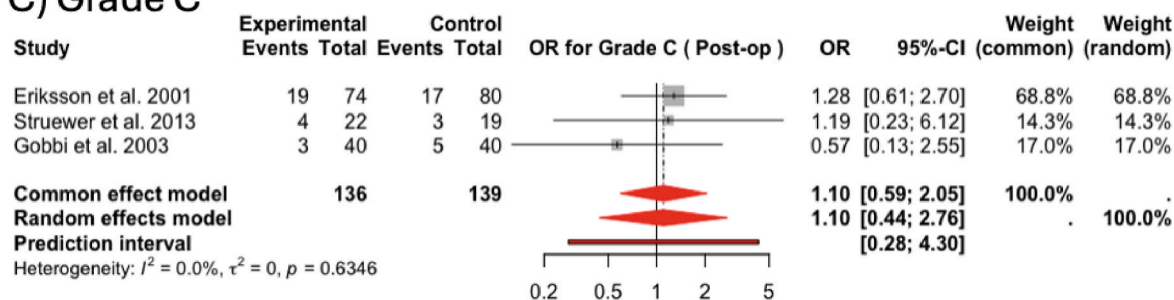
A) Grade A



B) Grade B



C) Grade C



D) Grade D

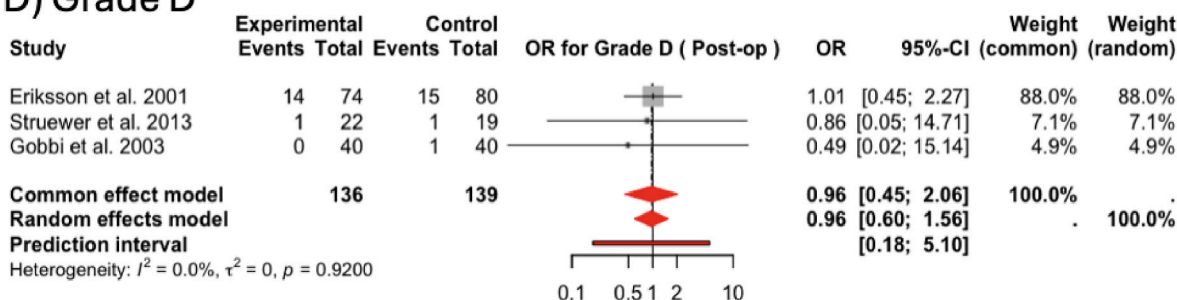


Fig. 6. Post-op forest plots for IKDC Grade A–D scores.

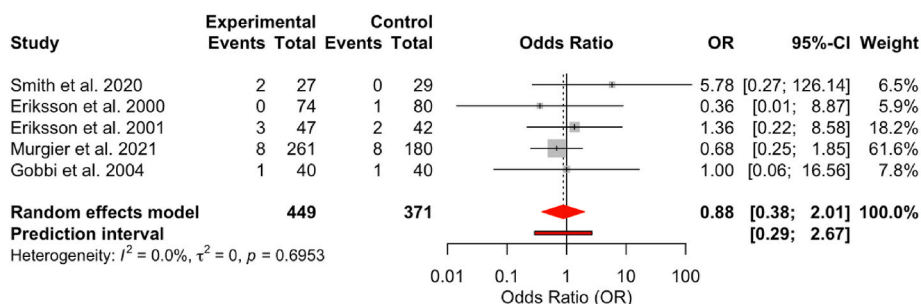


Fig. 7. Post-op forest plots for graft rupture of both ST and BTB grafts.

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Appendix 1

MeSH Search Terms

((1. “ACL.mp.”) (2. “Anterior cruciate ligament.mp.”) (3. 1 OR 2) (4. “BTB.mp.”) (5. “BPTB.mp.”) (6. “bone tendon bone.mp.”) (7. “bone-tendon bone.mp.”) (8. “bone-patellar tendon-bone.mp.”) (9. “bone-patella tendon-bone.mp.”) (10. “bone patella tendon bone.mp.”) (11. “bone patella tendon bone.mp.”) (12. 4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10 OR 11) (13. “semitendinosus.mp.”) (14. “hamstring autograft.mp.”) (15. “hamstring.mp.”) (16. 13 OR 14 OR 15) (17. 3 AND 12 AND 16)).

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